Task 1: College Equipment Database

Objective. Create and query a database containing one table using SQL Server.

Scenario. A college laboratory manager needs to record equipment and identify which devices are available and where they are located. Each row represents one physical device. Two devices may have the same name, but each has a different identifier.

Database name: `CollegeEquipment` **Table name:** `Equipment`

Table definition

Column	Data type	Constraint
EquipmentId	INT	PRIMARY KEY
EquipmentName	NVARCHAR(60)	NOT NULL
Room	NVARCHAR(10)	NOT NULL
IsAvailable	BIT	NOT NULL

Records to insert

EquipmentId	EquipmentName	Room	IsAvailable
1	Laptop	Lab1	1
2	Laptop	Lab1	0
3	Projector	Lab2	1
4	Router	Lab2	0
5	Printer	Lab1	1

`IsAvailable = 1` means available; `0` means currently unavailable.

Required work

1. Create the database using `CREATE DATABASE`.
2. Select the database using `USE`.
3. Create the `Equipment` table with the columns, types and constraints shown above.
4. Insert the five records. Specify the column names explicitly and match the order of the values to the column list.
5. Retrieve all columns and all rows from the table.
6. Retrieve only the equipment name and room for every device.
7. Retrieve the identifier, name and room of every available device.
8. Retrieve the identifier, name and room of devices that are both available and located in `Lab1`.

Optional extension. Retrieve all records ordered by equipment name, then by identifier when names are equal.

Check your reasoning. Predict the rows before executing each query. Explain why device 2 should not appear in the result of requirement 8.

Working instructions. Execute one statement at a time and run each `INSERT` only once. Preserve existing work; consult the instructor if the database already exists. This task does not require `UPDATE` or `DELETE`.